

2022 Active Inference Livestream Review

ReviewStream | 1:56:19

Hello and welcome everyone. It's December 31st, 2022. We are here in the 2022 Active Inference livestream review. Welcome to the Active Inference Institute. We're a participatory online institute that is communicating, learning, and practicing applied active inference. This is a recorded and an archived livestream, so please provide us with feedback so we can improve our work. All backgrounds and perspectives are welcome. We'll be following good video etiquette for livestreams. Head over to activeinference.org to learn more about getting involved in livestreams or various other projects and learning groups at the institute. Well, today is going to be great and fun. We are simply reviewing our streams from this past year. We're going to touch lightly or mildly on all the streams that we carried out from 2022. And by way of doing that, we may end up reflecting and pre-flecting, thinking about where we're going to head in 2023. So we'll begin with just an introduction and a hello. I'm Daniel. I am a researcher in California, and I'm excited to reflect. This is like the .0 preparation, .1, .2 discussion, and then afterwards we reflect.

This is like the bookend reflection for all of those. And so it's like an even deeper level of nesting when we're going to juxtapose so many things that we talked about. And I'm sure at some neural level, whether somebody was familiar with the streams or whether they listened to none of them, there's going to be something about just touching on so many topics and so many perspectives being juxtaposed that I think is going to be really pleasant. To you, Dean.

Yeah. Thanks, Daniel. I'm Dean. I'm here in Calgary. Maybe I'm a revised or recycled researcher. I don't know what my title is now, but after Daniel said, yeah, I think we should do this again, I was like, yeah, we absolutely should because it's, my metaphor is that it's kind of like double, double frying those French fries. It's a, it just adds an extra layer of integration or crispiness. That's a hard to resist. You, you get a, because the way that we present these things is, is hopefully a way of integrating a bunch of information because we talk amongst ourselves and it's helpful. And then you look back at yourself talking about yourself and others, and you just get a whole different perspective on it. So I actually, I'm really looking forward to sort of running through these things. It's going to be a sprint, but that's okay. It still gives you a chance to figure out what the milestones were. It's like seeing the entire Tour de France in, in a 30 seconds or 30 minutes, which is kind of fun.

It's like we completed that marathon and now we're high speed watching the recording from our body camera. All right. Well, we'll keep with some patterns from the more paper centric discussions, like leading with big questions. So what was our stream of attention in 2022 and where and how will we stream in 2023? What did you have to add on these questions?

Not, not, not a heck of a lot. I mean, it's, it'll be interesting given the amount of time that you committed to the endeavor of here's 2022. I wonder what we'll fill it with. I'm curious now that now that you're on the backside of that, what parts of what you hoped would be achieved came about and

what parts do you think are still, there's still room for some, some backfilling?

We'll come to that after we've touched on everything.

Yeah.

Yeah. I really liked what you added here though, in a world that is becoming more familiar with congregation and digital mediums and spaces is our use of technology more than an easy way to gain information access. You can be in the library all you want and you won't read a book. We can have all the access to information in the world and we won't necessarily digest and incorporate. We can have all the generative AI in the world writing in these amazing essays and making art, but where does that put us as information forager? And then you added, can we turn figuring out into out there figuring? And as more and more people start to want to figure out active inference, it's not a Rubik's cube on the table that gets finished. And by putting it out there figuring is where active inference are.

So I think that's really exciting. Like we do have so much to figure out from all backgrounds and yet we want to be out there figuring and having impact.

I think that's where your background in field research and my background in sending people out into professional settings, very different, but also structurally very similar. There's no substitute for actually picking yourself up and it's a form of migration. I don't, I, you like to use the word stigma to T until I can actually figure out what my pheromone trail is. I'll, I'll let you have that piece of it. But I mean, I, I understand it. I get it. And yeah, let's see where it takes us.

All right. Roadmap. So maybe we should have done like a, a, a rear view mirror or something, but we had 87 streams in 2022. The live streams table has 87 rows. So it was an amazing year and the numbers are shown here. We're going to touch pretty lightly on all of the streams from two through nine, the round table, symposium, math stream, org stream, Twitter spaces, guest stream, model stream, and book stream. And then

we're going to slow down a little bit and talk about some of the papers that we discussed.

So first the overall review though, for a general Institute level review, check out our quarterly round table number four. So that's where we talk more about the Institute's directions, other projects and learning groups. But today we're just going to talk about the live streams. All right. What did you write with these questions? Yeah. I'm, I'm, I'm, I'm often curious because when you're, when you're busy constructing 87 rows, one, when, when you reached, whatever that limit is, is there a big picture that has come about as a result of that stack? Again, it'll probably be something we'll be able to answer better at the conclusion of our, of our chat today. But I think it's something to try to keep in mind. I don't know that we can say with certainty that you can keep your big picture in my sense of that same picture is the same interpretation, but it's, it's a good, it's a good idea to keep a sort of a general level sense of, so did this all come together or, you know, how did the live streams intermingle with some of the guest streams intermingle with some of the book stream stuff that you, and is it Tyler and, and Blue are, are looking at right now? Yeah. Yeah. Well, the links here can all be accessed through activeinference.org. And of course, one project of note is the active inference journal, where all of the live streams through the incredible

work of Dave Douglas and others get transcribed, archived, and versioned up on the GitHub. So that is one structural development of the live streams. Is there a big picture? It really can feel like just moving pebbles back and forth. And there isn't necessarily a vision of what the final nest looks like. So I hope we can kind of look back on it once we've gone through it. All right, let's jump in. Yeah. First round tables. So we have a tradition of four quarterly round tables at the Institute scale. The first two, one we'll notice, are act-inf lab quarterly round tables. And the second, or the third and the fourth are active inference Institute round tables, because in the middle of 2022, we had, of course, our transition from being active inference lab to active inference Institute. So here's an image meme that we have from the fourth quarterly round table. And these round tables, the screenshots with who's there and who shows up, but also just what we discussed are like planting a series of flags in the ground. And they're an absolutely honest representation of who we were and what we were doing then. And so I really enjoy sometimes even looking back and seeing how things have happened. So that's not as scientifically or philosophically conceptual of a live stream series. And the information in certain ways is out of date because it's more of an informative participation oriented communication, but nonetheless, a great series. And also before we head in and to avert saying this of every single series and stream, big appreciation for those who had the bravery and capacity to join, whether they joined repeatedly or one time, it's been awesome. And it's totally co-created with the people who are there in the most just unbelievable ways. All right. To the symposium. So this was our second applied active inference symposium. The focus was on robotics and the first applied active inference symposium in 21 was just with Carl Friston. It was a one man show. And for this symposium, we had some more elaborate developments. We had some external co-organizers in Matt Brown and in Mark Miller. We also wanted to host a wide variety of speakers and perspectives. So we had two four hour sections with one, two, three, four, five presenters in the first session and one, two, three, four and a half with Friston joining in the second session. And we saw the state of the art in active inference being applied to robotics. We saw embodied actual robots. We learned about some of the theory. We saw connections to previous lineages of robotics and also more general control theoretic areas. Each of the sessions closed with a round table discussion. The second one being more populous than the first. And when people say like, okay, enough about what active inference is, how, and where can it be applied. And people commonly point to areas like human health, as well as robotics as some of the leading areas in the recent and in the coming years. And I'm just really happy that we were able to highlight the state of the art. Like this is what robotics and active inference looked like in 2022, publish a transcript, make it citable and give visibility across these different sectors so that they could connect on some of their developments and also their ongoing challenges. Mathstream. Yeah, go ahead. Go ahead. I hope that that at some point, and I don't know, timing really matters, but I hope at some point people in other more dedicated learning spaces like formal education spaces systems will find themselves asking questions around how things that we've been talking about and we've been interacting on might complement the way that things happen in those systems. So it's not, let's go build a robot that does X. It's actually

taking the active and the inferred and taking actions on hidden states as a complement to, you know, we've got to

get a certain score on an SAT. I think it would be very interesting to see when that, that coming together on the horizon, we know that the

train tracks remain parallel, but as we look at F, there's a name for that, you probably know what it is, but when it reaches that horizon, it appears they've come together.

So I'm hoping that that isn't sort of lost in some of these other initiatives that I know would actually be easier to do because there isn't such a history and a whole bunch of people who are very vested in maintaining the status quo as opposed to really journeying in ways that are not discarding what we know, taking the priors and allowing for some new priorities.

So I just want to kind of toss that in there.

Awesome. Thank you. We had just one math stream formally with David Spivak and Blue facilitating on category theory. This is a few train stops away for some of us individually and for the field. But this year, there have been some incredible advances in active inference and categories.

So the field is being fertilized and prepared for an incredible regenerative category theoretic implementation of active inference. But that's all we'll say for now here.

In the org streams, we focus on the intersection of active inference and organizations. We had two streams.

First one was with Eugene Leventhal from the Smart Contract Research Forum. And the second stream was with Bijan Hezri, whose book *Governing Continuous Transformation* is also the focus of the book stream, which we'll get to in a second.

And both of these were awesome streams. There are hidden states for both. I'll only say what I'll say, which is that they both were incredibly informative. And the triangulation of active inference for both of them was something that was incredibly natural.

natural. Coming from, as in Bijan's case, somebody with decades of executive experience who completed his PhD dissertation on free energy governance and sees the framework specifically as a vehicle for moving the frontier of leadership and organizations.

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conversation.

And we connected a lot of.

topics of interest in.

online and web three organization.

So it was a great.

time.

Yes, something that if you.

want to do something interesting, watch the work stream with.

Jan and then watch the live stream series number 36 about modeling ourselves.

And you'll very quickly see that.

Jan was closer to the end of the year, but we were many of the things that he spoke of.

We were talking about in detail.

With one of the papers that we took up on basically a year ago.

And that's the interesting that you can't, you don't realize that until you're on a day like today.

Awesome.

We had one Twitter spaces.

The title was can web three survive without cognitive modeling.

It was a provocative.

Clickbait headline.

And we had.

A chance on this discussion.

To talk about the active block friends project.

With Yacob here as one of the primary developers and advocates and stewards of that project.

And we discussed this question, which is can web three or cyber physical systems more broadly exist in an increasingly complex present and future without something like cognitive modeling for which we believe active inference is the most composable and flexible, generalizable and powerful framework.

So I forget whether it's never ask a question in the headline if the answer is yes or no.

I always forget.

So I never know.

So I always approach those headline based questions with an open mind.

But we had a good time in this discussion and these kinds of informal formats might have a lot of applicability.

We didn't really need to prepare much.

There were also weren't any slides.

So for those who are interested in this genre of social audio or whatever it is called, there's a lot of opportunities for frankly, just interacting with people where they are talking about active inference and making the compromise in the improvised moment.

All right.

All right.

All right.

We had quite a handful of guest streams from number 14 to number 31.

So I'm going to just briefly try to recall and mention those who joined.

Jorge Estvist discussed osteopathy and mental health in 14.

Bobby Azarian in 15.1 and 0.2 discussed universal Bayesianism.

Mark Solms joined for 16 discussing consciousness as precision optimization, which also was published by the Active Inference Journal in one of the most manually crafted and beautiful publications.

Mal Alboroson spoke in guest stream 17 about epistemic communities under active inference, which is an area of increasing importance.

Ian Robertson, Michael Kirchhoff, and Julian Kiverstein discussed the literal literalist fallacy and the free energy principle, where that was only in March.

It was on Pi Day and we reawakened the kraken of the realism, instrumentalism, dialectic, which I know that we're going to touch on today again.

19, Marco Faccion discussed extended predictive minds.

In guest stream 20, Shannon Proksh discussed coordination dynamics of multi-agent interaction.

And the screenshot on the bottom left is showing a slide from her presentation.

I think it was an amazing presentation.

It's very accessible and very cutting edge work on multi-agent active inference.

And also in the very fun setting of music, listening and creation.

So it was very engaging.

Adam Saffron in guest room 21 discuss free will.

Anna Lemke in guest room 22 discussed dopamine nation.

And that's a fun one to view and share because Anna comes from a psychiatry background, not familiar with active inference too much, but open-hearted, open-minded and engaged with real people.

And with research every day.

And with research every day.

So we had a really affable conversation.

In guest room 23, Maxwell Ramstead and Dalton Sacktivadevelle joined for rebooting the free energy principle literature, which was a paper discussed.

That was not the title of the paper, but a paper was discussed.

They got folded into the pre-play for live stream 49.

So we'll talk more about Bayesian physics.

Then Steven Grossberg joined for guest room 24 explainable and reliable AI.

And also a commendation there to Ali, who did incredible preparation and facilitation as well as editing the transcript here and made this an incredibly high quality experience for everybody.

Guest room 25, Brett Anderson discussed the continuum of traits between autism and schizotypy.

So that was kind of interesting work and speaks to a lot of areas of meaning.

In guest room 26, along with my colleagues at Common Sense Makers, Daphna Shahaf, Lauren Hebert,

Ronan Tamari and William Fisher, we discussed from users to sense makers and about stigma G, which is collected behavioral algorithms that modify the environment in the context of sense making online.

Guest stream 27 was with John Verveke and we had a great time.

We had a stool with three legs, predictive processing, relevance, realization, active inference.

And Dean and I were preparing those slides.

We had the stool going and then we flipped the stool and it was upside down.

And then that was just one of many inversions and transformations.

John has a very delightful and mindful way of approaching dialogue as those who are familiar with his work know.

And so we had a great time.

In 28, Giovanni Rola discussed reconceiving rationality and inactive cognition.

Guest room 29, Shana Dobson on making up our minds, imaginative deconstruction in math art.

For those who are looking for the triple play of math, art and philosophy.

There is only one unicorn in this forest.

Guest room 30 with Kirtan Atreides was on the human governance problem, complex systems and the limits of human cognition.

This was a very fascinating presentation of an artificial intelligence system that was churning in the background while we had a discussion.

And so it was quite visually and conceptually interesting.

And then most recently in guest room 31, Brett Kagan and Adil Razzi discussed biological and synthetic intelligence from the free energy principle perspective.

And some of their empirical work with neurons in a dish playing video games and the way that they were modeling that with active inference.

So we had a great set of guest streams.

And we look forward to a diverse and comprehensive set for 2023.

So if you want to join for a guest stream or if you want to recommend or introduce somebody, this is very low bar.

It can be a presentation they've given somewhere else.

Some of these presenters requested us to do background, which we're happy to do.

Prepare questions.

Others just show up.

So if you think there's somebody who could fit in well in a guest stream, even if they're not like in the center of the active inference research developments ongoing, we can absolutely make it happen.

Dean, what would you add on the guest stream angle?

Well, if I could, if I can invite five people out myself at the same table and I could get Dalton and I could get Shauna yourself.

And John Verbacki and me just sitting around and just talking about stuff.

I think I probably would be about as good as it gets.

I think that would be just one of the most amazing dialogues.

And you wouldn't know where it was going to go.

And that would be half the excitement is just people who, who can, who can basically have a conversation with anybody about topics that are incredibly sophisticated.

And yet seem to be able to bring it to a level where, as you said, it's engaging.

It's, it's really, it's amazing.

And this is not to downplay all the other guests that were kind enough to join.

But again, if I were to, if I were to do something on a social level, on an informal level, all of these people, thank you all to all of them.

Great.

All right.

In the model streams, we like to highlight modeling.

And we had three absolutely cutting edge presentations with interactive components.

model stream five with Pietro Mazaglia and Tim Verbellien.

We discussed contrast of active inference, which uses some current threads in non-active inference machine learning, like contrast of learning and brought it into the context of active inference to facilitate learning in noisy environments.

In model stream six, Teofil Champion joined to discuss his quite prolific and creative avenue of research into branching time active inference.

And Jakob and Ali are here also did a lot of background to prepare.

And as we entertain larger and larger state spaces and deeper and deeper possible futures with more and more and more affordances and affordances on preferences and affordances on preferences.

And these state spaces become vast.

And things that you can write down on the back of an envelope might take all the heat in the world with current computers to run.

Just like you could write down two large numbers and multiply them, but you could look at each number and it's just a number.

So you can draw a base graph and yet to run that base graph might be not plausible.

So there's a lot of relevance in this area of research of finding out these computational acceleration methods and heuristics, as well as even just qualitatively, this idea of branching time, branching and pruning time, and what kinds of strategies are used to branch and prune time adaptively.

It's very exciting.

And then in model stream seven, Connor Heinz and Daphne Demeckis joined to discuss PyMDP, which is rapidly becoming the Python active inference implementation approach of choice in the community.

And it was great because we were able to ask basic questions.

We were able to hear about the intentions of the package and we're going to have a dot two in just a couple of weeks.

This is something in terms of the package.

It's something where more and more of our work in the Institute is going to use this package and the interoperability of different generative models and processes and simulations written in PyMDP is going to facilitate the accessibility of the

the rigor and applicability of the active inference.

So great work to all those who are contributors on this package and everybody who is out there applying active inference computationally and shared it with us this way.

All right.

And the last non paper live stream is the book stream.

So the book stream is being led by blue and Tyler Solberg.

And when they got a hand on Bajan Kezri's book, Governing Continuous Transformation, reframing the strategy governance conversation, they jumped on it.

And they said, how can we do some kind of a regular series?

How can we increase our accountability in reading, understanding this book and also make something for the community?

And they said, we want to do some kind of a weekly ish book stream.

So we've done that for the past several weeks, and there will be many more sessions to come, including discussions with the author.

So, great initiative.

And it's keeping the stove warm on these topics of organizational development, which are just absolutely evergreen.

All right.

On to the paper live streams, which we don't play favorites, but we love the paper live streams.

Go ahead, Dean.

Start us with 35.

Where were we in January 22 when we were reading A Tale of Two Architectures, Free Energy, Its Models and Modularity by Majid Benny?

I wish I could say that I looked a lot younger back then, but I looked old back then, too.

I think a year ago we were pretty excited because I think it felt as though things could go in a number of different directions and that we as a community that were discussing a lot of the most cutting edge on the sort of bow wave of information that was coming out of people's research.

It kind of felt like we weren't just aggregating.

We were actually kind of one step removed.

We weren't getting too locked into any one particular facet of this multifaceted idea.

And it could really blossom into something.

Like we had this really what we thought was this kind of magic acorn or unicorn or whatever, and how it would be viewed 360 some days later, we didn't really know.

But we also weren't afraid to kind of test the limits.

That's what I felt 22 was more than anything else.

It was an opportunity to test things, and that didn't mean that we were out to prove people wrong.

We just wanted there was not enough information and not a large enough sample size that we could do some cross checking.

And that sort of allowed the people that were doing the cross checking, the ones that were reading this long

list of papers, a chance to kind of go, huh, okay, well, I think I've seen that over here, but there seemed to be some hesitation around whether that how far that rule holds under this set of circumstances and yada yada.

So 35.0 started out, I've just put my notes here as a formal treatment of what graphical modeling can look like, i.e. our DAGs are cycle list while DCM are cycle based.

And for me, the introduction of a term joint probability up to that point, I'd never heard of that expression.

And we ended up using joint probability and a lot of the papers that followed.

So there was a bit of you could start weaving golden threads between things.

Majid, who eventually joined us for this, nothing but props for that guy for again, pulling in what are seemingly really distal concepts and making us realize, oh, well, if you do do that, you gain insight pretty quickly.

Majid, who we went on to 35.1, which was which do we start with representation and or effect both in order to model.

Majid, so recognizing the importance of partial information encapsulation that everything doesn't have to be absolute, especially over different time frames, not just architectures, although that term was in the title of his paper.

[illegible]

perspective swaps. I didn't realize that I had introduced that into the conversation then.

But yes, that's where it all was seated was back in January, almost a year ago. Then 35.2,

what are we trying to trace? Separation and or boundedness? The answer to that,

as Majid clearly concluded, depends on what's available. So availability, not just our regime

of attention. So I thought that was an incredible thing to bring up a year ago. Do we keep that?

What are what that we are the trans the tracers and the modelers and the separators, and the boundaries in the front of our minds as we analyze situations if learning is the sought sought after effect?

When does that help? Does this lead to a year of holding up when we are looking at a target system,

say active inference as subject? And when we as observers might be using active inference in

formal ways, i.e. for allowing priors to not get in the way of new insights, assuming insights are a

priority for agents. And I was interested in thinking, I wonder how they would be able to get

that into JS robots. Right? So that's an interesting thing. To me, this is the divide between being a

programmer of a robot where the priority is learned versus dictating how priors are amassed and organized

and then inhabiting a programming or participatory experience. One where deconstruction is adaptability

until ultimately something called complexity dominates the conversation. In both of those situations,

priors and priorities work in concert. But in the latter, the humanist cases are on opposite sides of a

Markov blanket. So at least partially hidden, but also partially encapsulated. We talked a lot about duck rabbits that are not riding two unicycles concurrently, whereas the robot might. That's an interesting difference there. And for humans riding two unicycles is equal and complete partitioning or encapsulation. So that's why it's hard to do. It takes much training to be able to carry that out.

So that's 35. That was Majid and bringing the science and the modeling and the abductive. And that was our conversation, even though his paper was in response to a couple of other authors years arguing that there's only one way really to model. And so again, lots of things grew out of a paper that was meant as a response and created a whole different ecosystem.

Awesome. Wow. I'll just give a short thought. Graphical models and network theory are at the heart of active inference as well as many other fields. And what it means to have a network or a graphical model is to represent a system or a model in terms of nodes and edges. And exactly what those nodes and edges are matters a lot. And in this diagram, we had the parable of Bob and Alice who brush their teeth at a similar time because people in that region just happened to do so. And we discussed how the statistical connections between their activities in terms of the functional and effective connectivity are not the same as what is touching what in the real world. And this very straightforward deflationary approach to separating the map from the territory came back again and again and again. It was like we were on the carousel of functional and effective connectivity and the map territory and the math is not the territory.

All of those topics are so central because when we're looking at those generative models, we're looking at a graphical map, not a picture of the territory. And it's easy even when you know to forget. And again, I often wonder how, because Daniel reminds me, you know, people vote which papers we're going

to do and sometimes it's just availability of authors. But the next paper, which was 36, could not have been a better co what's the word co inhabitant the with with Majid's work. It was like the perfect follow up math, Matt and Giovanni's work on on modeling ourselves.

Yep, give an overview, go for it.

Daniel Jameson-36 was the importance of working backwards.

Daniel Jameson-36 was the way that you and Blue set up the entire paper was well we're going to go to the end and we're going to come, we're going to work our way back through this paper just so that it makes sense because I think that there was a bit of a woo woo kind of

Daniel Jameson-36 sense from reading the paper because it was very, very deeply philosophical.

Also just introduced the claim that in order for internal representations to be wiped out, we need to marginalize the unit call inaction really radical. As if adding an exclamation point to your claim somehow changes the relationship of an action which is a part to the entirety, which is a whole.

Daniel Jameson-36 was the goal. So you when you guys first picked up paper and you're doing the dot zero.

Daniel Jameson-36 already there was like Okay, well, I wonder what we're getting into here.

Daniel Jameson-36 well you also asked indirectly how artists.

Daniel Jameson-36 indirectly how artists people with good external rep who are good external external representation lists. Maybe have a different gapping or a way of entangling or both.

Daniel Jameson-36 the organizational structural and content related and functional aspects

Daniel Jameson-36 is either representative or non representative within parameters.

Daniel Jameson-36 yeah nice table there set up which boy you got a lot of shelf life out of it.

Daniel Jameson-36 it was useful for all three live streams.

Daniel Jameson-36 what I found most interesting was that to be an activist meant to you the use of

Daniel Jameson-36 simple heuristics means control mechanisms do not need to get to the level of sophistication more than what.

Daniel Jameson-36 what short catch and long yet a great diagram of the baseball Carl hitting it out of the park.

Daniel Jameson-36 categorizing the FEP equals you can be all over the map.

Daniel Jameson-36 does that mean we can engineer a bridge to everywhere, which was one of the questions.

Daniel Jameson-36 I'd written now and actually after watching you and blue do the dot zero.

Daniel Jameson-36 then in the dot one.

Daniel Jameson-36 how does transition.

Daniel Jameson-36 go from materialized to non materialized and then back and that was.

Daniel Jameson-36 that brought us to the idea of what's stable and what's dynamic juggled as a process of predictivity.

Daniel Jameson-36 in order for representation and non representation to coexist this was the great thing they are they are bedfellows.

Daniel Jameson-36 Dean and blue debated the scale freeness and the scale friendliness of science.

Daniel Jameson-36 i.e observations recordings analysis and prediction.

Daniel Jameson-36 So modeling ourselves in real time with Daniel's substituting disciplinary for generalization as modeler strategy, the question was raised around what must exist pre plan pre model.

Daniel Jameson-36 And then we I don't know the lady's name who joined us for the.

Daniel Daniel.

Daniel Jameson-36 We got into some really interesting loops with your table.

Daniel Jameson-36 Again, we started we were looking at organizational structure.

Daniel Jameson-36 and content related as probably more stable things that we want to anchor with those edges and nodes represented and not kind of moving around all over the place or.

Daniel Jameson-36 Without edges.

Daniel Jameson-36 You can have a gap like I think there was a point that came up was the black lines separating the rows and.

Daniel Jameson-36 And the and the columns were they solid or were they actual gaps.

Daniel Jameson-36 So that was kind of interesting.

Daniel Jameson-36 And then there was a functional as radical dynamic and dependent on those stabilities.

Daniel Jameson-36 Now is modeling ourselves how when where what what if and which if we act.

Daniel Jameson-36 How does that impact the model that we believe exists as we boundary cross.

Daniel Jameson-36 Which again, if you take it to the stuff that vijan is talking about.

Daniel Jameson-36 On a on an organizational level.

Daniel Jameson-36 That gets.

Daniel Jameson-36 That's combinatorial combinatorial exclusive.

Daniel Jameson-36 So it's interesting that the how we model ourselves.

Daniel Jameson-36 Has a huge impact on how do we model ourselves as an organization.

Daniel Jameson-36 As a weave that which we'd like to remain constant with that which we'd like to see changed.

Daniel Jameson-36 So that was modeling ourselves and again.

Daniel Jameson-36 And again, it was math and Giovanni didn't show up, but man,

Daniel Jameson-36 Would have been awesome to have them there and kind of lead that circus, because

Daniel Jameson-36 It was great.

Daniel Jameson-36 It was great debate between.

Daniel Jameson-36 Blue and I because she really wants science to be seen as it and

Daniel Jameson-36 And I really like science to be seen as specific and you were the great moderator and said,

Daniel Jameson-36 No, it's both.

Daniel Jameson-36 So I was 36, modeling ourselves.

Daniel Jameson-36 Wow.

Daniel Jameson-36 Yeah, I totally forgot about

Carl.

Daniel Jameson-36 Hitting it out of the park.

Daniel Jameson-36 About Carl Bonds.

Daniel Jameson-36 Fantastic.

Daniel Jameson-36 Hitting the particle out of the park and about where is representation.

Daniel Jameson-36 Oh, representation.

Daniel Jameson-36 Where art thy sting.

Daniel Jameson-36 Yeah.

Daniel Jameson-36 It was too fun.

Daniel Jameson-36 It was too fun.

Daniel Jameson-36 And also just so people know, like the live stream page has the slides for every

Daniel Jameson-36 Every single live stream that we're discussing.

Daniel Jameson-36 You can find the live stream, the title, the paper, the transcript, the GitHub,

Daniel Jameson-36 Where you can contribute to the journal as well as the slides.

Daniel Jameson-36 So when we're pulling slides, you can also look at them.

Daniel Jameson-36 So yeah, this was 36 was awesome.

Daniel Jameson-36 All right, 37.

Daniel Jameson-37 Free Energy, A User's Guide by Stephen Francis Mann, Ross Payne and Michael Kirchhoff from 2021.

Daniel Jameson-36 You want to take a first pass?

Daniel Jameson-36 Well, I just remember that as much as I'd like to be able to say I could do the math,

Daniel Jameson-36 If somebody else does the math and I can follow along at my own pace.

Daniel Jameson-36 This is where I learned.

Daniel Jameson-36

Daniel Jameson-36 Yakub's gift, basically.

Daniel Jameson-36 So that's what my first pass is.

Daniel Jameson-36 I know when I know when I'm in the presence of people who could go

Daniel Jameson-36 A lot further on that plane than I can.

Daniel Jameson-36 So yeah.

Daniel Jameson-36 Yeah, the paper is an excellent user's guide or introduction.

Daniel Jameson-36 One memorable section was on this trilemma or trilectic of general or philosophical claims,

Daniel Jameson-36 mathematical claims and empirical claims.

Daniel Jameson-36 And they went through edge by edge and they asked,

Daniel Jameson-36 How do claims from each of those three vertices

Daniel Jameson-36 justify each other?

Daniel Jameson-36 And so we hear that all the time people speaking about active inference in other areas.

Daniel Jameson-36 Well, because the brain is this way, we use this equation.

Daniel Jameson-36 Or we use this equation because the brain is this way.

Daniel Jameson-36 Or because of this philosophical belief, we use this equation.

Daniel Jameson-36 It's like seeing it laid out that way brings a huge amount of rigor.

Daniel Jameson-36 And it also helps us understand that rigor is not just increasingly technical math.

Daniel Jameson-36 Rigor includes the relationships among mathematical, philosophical, and empirical claims

Daniel Jameson-36 So that was one key takeaway for me.

Daniel Jameson-36 And another was this, the three piece,

Daniel Jameson-36 The Tale of Three Ps.

Daniel Jameson-36 And it was a focus of the paper to point to the unconventional way

Daniel Jameson-36 In which preferences are utilized in active inference,

Daniel Jameson-36 Which is not as a reward or a goal function,

Daniel Jameson-36 But rather something that is kind of the fundamental frequency of the generative model.

Daniel Jameson-36 And so you have entities striving to return to the bottom of the bowl

Daniel Jameson-36 In terms of the path of least action, in terms of the frequency that they want to be at,

Daniel Jameson-36 And that is the embodiment of preference.

Daniel Jameson-36 And as we came to see later, homeostasis and allostasis.

Daniel Jameson-36 But there isn't some sort of dual track between what is and what should be.

Daniel Jameson-36 Rather, there's a bowl that represents what could be and is and should be

Daniel Jameson-36 Through its shape.

Daniel Jameson-36 And it's about the ball moving on that shape by which preferences come to be enacted.

Daniel Jameson-36 So that was something that was pointed to and very approachable.

Daniel Jameson-36 And they talked a lot about how they situated the audience of the work

Daniel Jameson-36 In terms of wanting to speak to philosophers who really did want to learn about

Daniel Jameson-36 What free energy principle is about.

Daniel Jameson-36 One of the things that I got kind of reinforced in my mind is because I kind of believe that

Daniel Jameson-36 Priors are on one side of a mark off blanket and priorities are on another.

Daniel Jameson-36 What that tale of three piece kind of in my head

Daniel Jameson-36 gave me a little bit of latitude on was preferences can be on either side of the blanket

Daniel Jameson-36 They can literally cross through they can they can pair up preferences can pair up with your priors.

Daniel Jameson-36 You can figure out what those things are that have been sort of burned into your your neuro neuro.

Daniel Jameson-36 Desire lines and they can also preferences can also be working with with your your priorities in this moment right as you find yourself.

Daniel Jameson-36 Now Okay, this is what the niches the information the niches sharing with me.

Daniel Jameson-36 But what's my priority.

Daniel Jameson-36 What's my preference as well.

Daniel Jameson-36 So that was what the three piece thing I remember now when we were talking about that is going, okay,

don't get the three things mixed up give each one of them their their time in the spotlight.

Daniel Jameson-36 All right, live stream 38 was evolution of brain architectures for predictive coding and active

Daniel Jameson-36 Inference by Giovanni Pizzullo Thomas Parr Karl Fursten from December 2021.

Daniel Jameson-36 And the topic here is so important.

Daniel Jameson-36 What is the evolutionary neurophysiological basis of cognition and how to complex cognitive phenotypes arise

Daniel Jameson-36 At a developmental timescale at an ecological timescale and at an evolutionary

Daniel Jameson-36 And where this paper eventually takes us

Daniel Jameson-36 Is into thinking about the graphical models again remember what we discussed just a few

Daniel Jameson-36 slides ago about graphical models.

Daniel Jameson-36 The graphical models which enable or facilitate or underlie different kinds of functional outcomes.

Daniel Jameson-36 And then how structural modification on graphical model architectures

Daniel Jameson-36 Enable different traits to emerge diverge duplicate and innovate.

Daniel Jameson-36 And so here we see on the right side it's a tree of options starting from a relatively simple initial graphical model

Daniel Jameson-36 How different kinds of modifications, duplications, etc. can result in the kinds of functional as well as structural

Daniel Jameson-36 phenomena that we observe in living creatures today.

Daniel Jameson-36 And whether that's the redundancy of gene regulatory networks

Daniel Jameson-36 Or the paralyzed architecture of the mammalian cortex

Daniel Jameson-36 These are the kinds of phenomena that this paper set out to

Daniel Jameson-36 Bridge the gap with between simply representing things as they are as graphical models

Daniel Jameson-36 And understanding what if you weren't just studying the ant brain today

Daniel Jameson-36 But what if you were studying the ant brain

Daniel Jameson-36 Over its phylogenetic history

Daniel Jameson-36 How would those generative models be changing?

Daniel Jameson-36 How do things like temporal depth or intermodal

Daniel Jameson-36 Come into play?

Daniel Jameson-36 And so that was a

Daniel Jameson-36 Nice paper

Daniel Jameson-36 That connects to evolution and

Daniel Jameson-36 A bunch of other behavioral areas

Daniel Jameson-36 Really helped me understand or confirmed in my mind

Daniel Jameson-36 That there's a big difference between a person who makes a model

Daniel Jameson-36 And somebody who has to interpret somebody third party model

Daniel Jameson-36 Because I remember going through this I was I felt like I was learning on the fly because

Daniel Jameson-36 Even as I was reading the paper, there was a lot there to try to parse and just having to be able to converse

Daniel Jameson-36 With you and blue and I think Stephen was a part of this one as well for a bit of time

Daniel Jameson-36 It made it a lot easier for me to kind of understand as an interpreter of models

Daniel Jameson-36 As opposed to somebody who's here's my model, what do you think of it?

Daniel Jameson-36 It was taking some of these really complicated models that other people have designed and then try to

Daniel Jameson-36 Make it into something that I had adopted

Daniel Jameson-36 Nice

Daniel Jameson-36 Okay, Livestream 39

Daniel Jameson-36 Morphogenesis as Bayesian Inference

A Variational Approach to Pattern, Formation and Control in Complex Biological Systems

Daniel Jameson-36 By Franz Kuchlin, Karl Fristen, Georgi Georgiev, and Michael Levin

Daniel Jameson-36 This paper took what in many other situations are mathematics of structurally stable but dynamically signaling architectures like the brain and considered the case of structurally volatile dynamically signaling structures. And so they focused on one of the key model systems in the Levin lab, which is the planarian morphogenesis. And upon cutting this animal, it quickly reforms into a self-similar body plan. So that has implications that are being developed in a whole host of areas related to health and development and disease. And the paper uses a

a somewhat technical, variational, Bayesian mathematics, which we also jumped into.

And we had Franz join us

Franz here in the office overlooking some gray featureless landscape and had just incredible insights to share.

And we talked about the environment, what are developmental priors and trajectories, what are preferences over morphologies.

And this really opened up the discussion from, as mentioned at the beginning, systems that have a structural stasis that do signaling in that structural space stasis.

All the computers are hooked up and now they're talking to each other, but they're hooked up the way they're hooked up.

Well, what if the computers can change how they're hooked up or what if they're moving around?

That is another space that also connects us to a lot of the physics of the embodied world.

Maybe there's friction. Maybe there's viscosity. Maybe there's a razor that splits two halves from each other.

And it was a great discussion to see how the mathematics of the variational Bayesian approach is able to accommodate

these different focal systems, which speaks to the generalizability of active inference.

There were a lot of formalisms there, but it didn't feel formal talking with Frank.

And it just felt like the possible with regeneration had a whole bunch of fields that yet to be explored.

And the kind of openness, thinking openness that opens up possibility was really, that was really cool about that live stream set.

Awesome. All right. 40. Who could forget?

A free energy principle for generic quantum systems.

2021 paper. In fact, a year and a day ago by Chris Fields,

Carl Friston, James Glazebrook, and Michael Levin. So this paper, we needed to situate ourself.

On one hand here represented as the X axis. Sometimes we find ourselves going back and forth between more

talking about free energy principle and active inference, other times discussing literature or ideas that are outside of the free energy principle. And then, orthogonally, we can think about this continuum between not quantum and quantum perspectives from physics. And we were in the upper right quadrant.

Or another way of saying that is we were everywhere because we were quantum and non-quantum.

And we were in FEP. We had the authors join for both of the streams, the 40.1 and 40.2.

So here we are with blue and Steven, us and the authors and every sequence of symbols that these fellows added was immense.

And to be able to make that daredevil jump across the Grand Canyon and connect the most advanced perspectives on quantum mechanics, quantum physics, quantum computation, unconventional computation more broadly today.

For example, thinking about the perception, action and memory phenomena in terms of repeated

measurements

of a quantum observer with their reference frame and looking at the holograph.

It was a great and memorable sequence.

It literally materialized the bridge to everywhere being in that upper right quadrant,

which is a fascinating thing because I don't know how you engineer a bridge to everywhere, but you, the possibility means you have to open up to more than somewhere.

And I think that that was again, one of those, one of those sets where you're kind of going, am I really doing?

Well, you wouldn't say this, but I would say this.

Am I really doing this?

Am I really having a conversation?

Do I get to sit at the same room with people of this capacity?

It was really fantastic.

And I got to talk to Michael Levin for privately after that because yeah, the idea of prediction matter expertise intrigued him.

So that was very cool.

Great.

Wow.

All right.

Carrying through 41.

Extended active inference, constructing predictive cognition beyond skulls.

By Axel Constant, Andy Clark, Michael Kirchhoff and Carl Friston.

So here we had, um, we had the all-star team.

We had blue, you and I trust is like glue.

If you don't have it, you won't be able to stick together.

And our team talked about trust and active inference models.

And I remember this section of the discussion where we had two systems, the blue and the orange.

Yep.

And we discussed like, what is their border?

What is their interface?

Is it another region?

But then if it's another region, isn't there a region that borders that?

And what is on the outside of that?

And so how does all that connect to extended cognitive systems extended relative to what?

And so this topic of extended inference was dealt with thoroughly here.

And the mathematics and the applications of extended cognition are so promising.

I also remember you discussing the difference between a type of active versus a passive offloading.

There's a difference between somebody writing a code on a post-it note that only they know.

So if they look at it later, it's like it helps them remember what they were thinking of.

But to somebody else, it doesn't mean anything.

And it doesn't do anything as just notes on paper versus JF creating a robot and sending it into the environment.

Yes, it might still need recharging and so on.

So it's not like it's totally disconnected, but it's doing things actively.

And that kind of active or passive offloading and the way that that facilitates our cognitive goals was very rich discussion.

And Axel, being the first author of this, he always, to me, he's like Majid.

He brings in the influence and the how long do you stick with something before you no longer, you know, can carry something on.

What is that?

He's such a good explainer of you realize that the environment is actively inferring you as much as you are inferring it.

That's really hard for people to get their minds wrapped.

I mean, the first time you hear that, you're kind of like, what?

But he does such an amazing job of explaining that.

It doesn't it doesn't seem like people talking about some silly thing like fairies dancing on the head of a pin.

Yep.

You actually go, oh, wait a second.

You're right.

It would have to.

I would have the environment would have to be scrutinizing me as much as I'm scrutinizing and trying to figure it out.

Nice.

All right.

42.

Robot Navigation as Hierarchical Active Inference by Ozan Katal, Tim Verbellén, Toon Van de Meyley, Bart Dött, and Adam Safran.

This paper was an incredible example of applied active inference.

They, in this paper, represented the full rainbow from conceptualizing a generative model to implementing it and making a robot that actually enacts that generative model in a warehouse setting.

And what that warehouse setting required from the generative model was that there was two levels at which the generative model was playing out.

At the higher level, which we were thinking of in terms of strategy, the robot decides where it wants to be, where it prefers to be, and where it is.

At a lower level, which we connected to tactics is related to the kinds of bodily moves that it can engage in.

It's more movement affordances.

And this discussion facilitated a lot of action into the math, the computation, the embodiment, the robotics,

and gives some of the clearest examples of applied hierarchical modeling that came back again and again and again.

So that was fun.

I was collaborating with Maria on the dot zero and Maria's philosophical knowledge with predictive processing took us way back.

And that was rooting the discussion in the ancients and thinking about world knowledge traditions and Kant and Helmholtz.

We hear a lot more about that kind of 17, 18, 1900 philosophy time.

But this pulled us temporally back and helped bridge the math of this paper, which was technical, to the ways that people have simply perceived, cognized, acted, and thought about the world around them.

And then to speak to the math, this paper, almost the entire thing before something like the last several pages, it's about hierarchical predictive processing architectures, but there's no action to be found.

It's just about inference, inference going on.

When they're doing inference on action.

The consequences of action are taken into account.

Policy selection is treated as a formal thing to do inference on.

Then you get active inference.

And it really isn't too much more complex than that.

And then just a last point.

Just like we had the three areas of claims justifying each other from Livestream 37, with the general claims, mathematical and empirical claims, we had another triangle in 43.

So here we had on the top biological systems and what they were doing.

On the left, we had philosophy and qualitative claims.

And on the right, we had formal quantitative claims.

And so we made another triangle with pretty much the same three vertices, which is mathematics, philosophy, and empirical biological phenomena.

And we took another first principles pass on constructing those three areas.

And what are the nodes?

What are the edges?

And how are inference and action at the heart of all of it?

All right.

44.

44 was The Therapeutic Alliance as Active Inference, The Role of Therapeutic Touch and Synchrony from 2022 by Zona McParlin, Francisco Serratelli, Carl Friston, and George Estvis.

So this paper was awesome.

Also here from the screenshot of the event, we have Ian Tennant with the Innersense channel and project who saw this paper, brought it to our attention, said, I want to facilitate this discussion.

And I want to connect interoception and the therapeutic alliance in a modern context to active inference.

And I want to bring in these awesome authors and make this a conversation that really hits home.

And so thanks, Ian, for joining and helping that way.

The paper focuses on the therapeutic alliance, which is like the clinical or medical or therapeutic niche.

It's the people in the room.

It's the receptionist.

It's the nurses.

It's the surgeon.

It's the physical rehab practitioner.

It's the family of the patient.

It's the patient themselves.

It's the physical surroundings.

And to be able to very broadly discuss that vital setting while connecting it to key active inference topics like generative model, synchrony, and so on, as well as to see their delightful figures and the way that they represented the same.

A bunch of authors joining for both of the discussions.

And it was an awesome series.

All right.

This one.

45.

This is what I come back from the dead.

And this is what I walked into.

I knew there was a reason why I wasn't called away.

It was because of 45.

45 was the free energy principle made simpler but not too simple by Carl Friston, Lancelot DaCosta, Norsejíd, Connor Hines, Kyle Uhlolzerfer, Grigoris Pavliotis, and Thomas Parr.

Well, I think the best way to summarize or crystallize the experience of this stream was Blue and I spent many hours preparing for the dot zero.

We were having separate meetings and doing asynchronous work to really work through and understand the mathematics of this paper.

And we did the 45.0 in May.

It was all great.

We were looking forward to June.

We had invited the authors.

We didn't really know what was going to happen.

And it was several minutes after the hour where we were supposed to start.

And so we usually just try to start on the hour or a few minutes after the hour.

Why wait longer?

And we just didn't have anyone else on stream.

So we said, all right, this is it.

We floundered through the dot zero.

Let us swim again into the dot one.

Yes, please, Dean.

I'm sorry, but you're not telling the whole story.

When you are completely out of your niche and you guys worked so hard to make sure that you could adapt for this.

But then you took the courage and the strength to literally jump out of the ocean into the boat.

And then all you wanted was, come on, it would be wonderful if somebody could show up and at least tell us that jumping out of the ocean was a good thing to do.

And then just like that.

Yep.

Yeah, just a few minutes after we started.

Come along and say, hey, how's it going?

Here, catch and release.

It was so amazing.

And I mean, you may not believe that it was that big of a deal, but it was.

And it was amazing.

Yeah.

Just a few minutes after starting with the two of us.

First, Carl Friston joined.

Then several minutes later, Thomas joined.

So, yes, it was one moment, but it spoke to putting it out there.

Yeah.

And being ready to show up and being ready for others to show up.

But also for these two sessions that both Carl and Thomas joined for.

For those who want to know about the mathematics of active inference.

Yeah.

There's many miles to go with making it accessible, rigorous and applicable.

So, not to be denied.

But if somebody wants to know the absolute state of the art, these conversations, you're hearing it directly.

Yeah.

And the questions and the discourse are 100% approachable.

So, this was just some of the highest density learning and higher learning.

It was an institute of higher learning.

If you want evidence of naivete, which is not something you want to remain as all your life, but those moments when you allow yourself to be naive and how that can be a feature.

Watch this.

If you want to, if you want to first step, you might look at going, I don't know anything about active inference.

Just still.

So, you may not understand some of the things that were discussed and that doesn't matter.

What you really will appreciate is the generosity.

Yes.

Great.

All right.

46.

So, 46 was active inference models do not contradict folk psychology by Ryan Smith, Maxwell Ramstead and Alex Kiefer.

So, what is there to say?

We are folk psychology.

We are them.

And this is also a topic that people bring up all the time.

Oh, preference.

Are we talking about like that I prefer to have this happen in my life?

Belief.

Are we talking about like what it feels like to believe something?

And this paper just absolutely squarely targets those questions.

And the conclusion is in the title.

Yeah.

Or the authors.

It's not a reification of folk psychology.

It's not chapter two on folk psychology.

It's not a nullification.

It's not a contradiction.

It is a non-contradiction.

Right.

Which is exactly what it sounds like, which may infuriate those on any side of multiple fences.

So, it was a great conversation.

Because the topic was so directly tangible and accessible.

We did want to connect the math and formalism of active inference to how people talked about folk psychology, which is just like us day to day talking about our feelings.

So, that was an amazing context.

So, that was an amazing context.

And also, this sort of fun stream memory was I used social media to communicate that the dot zero was beginning.

I guess on June 14th here.

And Ryan Smith just says, hey, I don't know if I'm going to be free later, but I'm free now.

So, we had Ryan in the saddle for the dot zero, which was the first time we ever had an author join and bring

that kind of insight into the dot zero.

So, you know, class was held.

So, we were, Court was in session and Ryan has joined multiple times and has so much incredible clarity on the math and on the psychology and the applications of active inference.

Anyone studying his publication or presentation career can know that.

So, it was a great conversation with Ryan and then also with Alex Kiefer joining in the subsequent discussions.

And Dean, you brought so many fun images and motifs to bear.

It was just like, it was such a fun house.

Well, here's the thing.

And again, this was by June, right?

We'd started in January trying to figure out what do the lines between mean, right?

So, is active inference a divider?

Is active inference an enveloper?

What this paper said was it's not contradictory to appreciate that it could be both.

That if we have a circle, you can be inside with the spokes or you can be outside and be the contact service.

If it's a divider, it can be on either side of the line.

But if it is both, then why would you always feel that you're doing something wrong by holding up both, right?

Why is both the contradictory aspect of this?

Which, again, encouraged me and my Minto, I don't care how you want to do it.

You can collapse past that.

But if you want to get to fractalization, fractalization being the moment that you're literally in the gap or on the line or on the boundary, right?

That's what affords fractalization is the fact that you don't necessarily go, well, that's a contradiction.

Hands clean.

We're done.

No, I have to stick to something simpler because I just, I will not accept a contradictory space.

There are times for absolutisms, but there's also what Ryan was pointing out, which is that maybe that's a feature.

Maybe this ability to see both is something we should just not, you don't have to accept it.

But what, again, counterfactual, what if you did?

So it was really good.

And then Alex showing up afterwards, kind of playing cleanup.

It was really cool.

Yep.

So that was taking us to the end of June.

We then took a few weeks off of discussing papers.

And in that inter-semester break, we came to 47.

So I'll let you describe 47.

It was Active Inference and Abduction 2021 by Ativeko Pieterin and Majid Beni and an Active Dynamics.

Social Cognition and Active Inference by Ines Hippolito and Thomas Van Es, 2021.

Yeah.

Well, again, there was, I guess there was an intention on my part to get, first of all, muster up the courage, because these people have spent a lot of time putting these papers together.

But my sense of it was, I've got both eyes open.

I'm not focused on instrumentalism or inaction or taking a realist stance or taking a non-realist stance.

And yet there was all of this conversation around those things as to which one entails the other or which one came first.

And all I wanted to do was hold the papers up side by each.

And rather than having people having to defend one side or the other, I wanted to hold it up as, here's two sides to the same coin.

We are on the edge of this coin now looking down on both.

What does that imply?

That's essentially what that was about.

Because I think at the core or the heart of prediction matter expertise, not just active inference as something that we can subjectify, but that we can act upon in formal ways, not just informally.

That's when you know you've entered the divide or you're on the boundary, which leaves it up to you as the person now that's taken up those coordinates to do things like we're doing today.

Reflect back on what got you here.

Was it one leg or the other, right?

It's like saying, oh, well, it was my left leg that got me here.

Well, it was, but it was also the right leg.

So how do you square that in, you know, some of these debates where people, like I said, they invest a lot of time and energy and effort into trying to make sure that what they're saying is correct.

To the point where they almost feel like they have to defend something because it narrowly fits.

And I wouldn't argue against that.

But if we're talking about both scale free and scale friendly, which we are, I think we all kind of accepted that now.

So then how does that work?

And Majid, despite the fact that he was not under the most optimal health conditions, was super kind, showed up and once again graced us with trying to explain some things, despite the fact that I know he wasn't in the best of condition to do that.

So again, another shout out for Majid Benny and his willingness to go the extra mile.

Yep.

And for you in breaking the one paper per series speed limit and having us consider two papers wholeheartedly.

So that was hopefully it was helpful.

That's all I was hoping is that, you know, it wouldn't be seen as as being disrespectful because we weren't giving enough attention over to something.

We were actually entertaining availability with attention.

Again, we were men doing things.

And yeah, that was cool.

Yep.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

All right.

Getting into the last three papers here.

Yeah.

And the reason why I mentioned the gap between 46 and 47 was because we came back with a fury and we weren't mad at anyone.

We were angry, but we were furious.

And we had some incredible streams.

So almost center in representing that for me is in Livestream 49, a worked example of the Bayesian mechanics of classical objects by Dalton Socktevideville.

So for this paper, Ali Jakob and our colleague, we met weekly for about two months to prepare for this discussion.

Because this paper.

And more generally Dalton at all.

And more generally Dalton at all.

And more generally Dalton at all.

Advances in merely 2022 have transformed the status of active inference in the free energy principle.

If you want to learn more about what that means, check out Livestream 49.

Ali, Jakob and Ander put in a ton of work.

We worked as hard as we could leading up to the dot zero.

And then Dalton took us on a journey in the dot one and the dot two.

We couldn't have asked for a better elucidation or a more enthralling presentation of Bayesian physics.

I really mean it.

And whether you're working in classical thermodynamic or quantum mechanics or whether you're not familiar with those areas,

this direction of work and specifically the presentation and the answers that Dalton provides

and just the spirit that he brought are absolutely memorable.

So I'll just give you a little aside.

This is the only time on any live stream ever where instead of playing music when I'm doing my workout, my third listen, my third listen of this, of the dot one of 49.1 was while I was working out because it's that good.

I could act because it had to be my third listen because the first two listens, I was just like having to pick my jaw up off the floor.

But by the third time, I wanted to listen to it a third time.

And instead of listening to music, I was listening to you guys work through this stuff.

And I was just like, this path integral stuff changes everything.

Yeah, just briefly for the conceptual.

Here are the three faces of Bayesian mechanics using a system and a scheme that's presented in this research

and in earlier research by Dalton and Maxwell et al.

One setting is where a ball is resting on the ground and it's not moving.

That is kind of like a distribution, a belief distribution that's getting new input, but it's not updating.

It's just getting the exact information it expects.

It's not updating.

It's not moving.

In contrast, when new information is provided, the prior updates to the posterior and it moves to track the mode of the new optimum.

So that's like if you had a ball on a staircase, then you pulled one stair and it fell down to the next one.

And you pull the next stair and it fell down to the next one.

Those both represent taking densities over states, like preference as a state, observation as a state.

States can change through time, but they at any given time are state-based.

Yeah.

In contrast, and that's where we have the non-equilibrium steady state-ness density.

So all the discussions we had about non-equilibrium thermodynamics and non-equilibrium steady states, all of that nest density discussion is on the states.

What some of these recent developments in path integrals, gauge theory, caliber theory, maximum cal, as well as the dual, the discovery of Dalton about the duality between the constrained maximum entropy principle

and the free energy principle.

When we calculate densities over paths, there does not need to be a NESS assumption or a discussion of a mode,

but you can get path tracking, path of least action tracking, without appealing to this intermediated NESS.

It's only been a few months, but there's so much more to come here on the mathematics and on the applications.

It's just, it's incredible.

And a year ago, we couldn't have said any of this.

So a lot has happened in active inference, but in the 49.0, you'll see us grappling with the eras of active

inference and FEP

and about what phases and epochs.

And even now we can acknowledge that we're in a post-Deltonian era.

Yeah.

And it's funny because I, before hearing this, I, it almost felt like you had to go down the vertical and then across the horizontal

and then down the vertical and then across the horizontal.

And then all of a sudden he introduced the wheels and the bike went riding down the steps.

That's how different it, it, it became.

Now that's still a bumpy ride.

You have to go quite quickly to smooth things out.

Right.

Smoothing.

Right.

But that's, that's essentially how dramatic it was like, it was like introducing the wheel.

Yeah.

It's only, it's only a wheel, right?

Wheels within wheels.

Yeah.

All right.

50.

So in 50, the paper being discussed, I didn't bring it in, was interception as modeling, allostasis as control with Eli Sinesh and Jordan Thurow, who are here as well as several other authors, Karen Quigley and Lisa Feldman Barrett, maybe some others.

So what, what would you add on 50?

Well, just that, again, you would see that interoception would be its own field.

And allostasis as being a separate field.

And then there's this lab in somewhere in Northeastern United States, that's trying to de-field, bring things into a, an opportunity to be integrated.

And it was fascinating to try to see how people address operationally and formally things that seem to be pretty messy and hard to manage when we're talking about things like we don't have necessarily access to the information in a mathematical way.

It's a, it's called a feeling.

So how do we, how do we blend those things together?

And again, I, I'll be honest.

I was the most skeptical and became a fan boy after having great conversation with Jordan and Eli.

Yep.

That was a memorable discussions.

Eli gave a presentation that helped us really like lock in conceptually how they were framing it and what they did.

Some of the statistical transformations are evergreen in the sense that there was a capacity curve, a probability density implied by a capacity curve.

And then a responsiveness that was a log probability density.

And now whenever we see one of these, we see all of them.

Right.

Because they're all just transformations, either PDF to CDF or log probability densities.

Check out the stream to learn more.

But whenever we see these, we can kind of see all of them when we see one and see one thing, see multiple things, learn one thing, learn multiple things.

Yeah.

It was awesome.

All right.

And the final live stream discussion paper.

So this was a canonical neural networks perform active inference 2022 paper by Takuya Isamura, Hideaki Shimazaki, and Carl Friston.

And this was a fun and an honor and a learning journey for me.

I prepared the .0 myself, which hadn't happened for a little bit of time by then.

This was just in November.

And then in both the .1 and the .2, it was just Takuya and I.

And the paper's implications are so stunning in light of the prevalence of neural networks and artificial intelligence machine learning.

The paper supports a claim that was latent within an earlier research paper by the authors, but they kind of completed the loop in this paper to really reinforce it, that as per their figure one, every neural network, canonical neural network, listen to the live stream slash read the paper to learn more.

Every neural network in engagement with the environment through perception, cognition, and action can be understood using the particular partitioning that we use in active inference, where the generative process engages the generative model of an agent, also in terms of perception, cognition, and action.

In fact, the training by which agents can learn from their own experiences about what to do in the future is also analogous as well.

And here we see some message passing reflecting that the discussion was technical, that the discussion was technical, yet asked basic questions.

And everybody who works with neural networks or knows somebody who does,

I'd recommend that they check out this discussion in paper.

because it was a perfect example of like making the bridge.

And here we see like the min2 of min2s.

We see the partition between the generative model

and the generative process,

and then two views on that being juxtaposed.

And so now when we see a neural network,

we can think that there's a generative model

that might reflect the semantics

of what the neural network does,

even if the structure of the neural network is arcane.

And in contrast, when we see a generative model,

we can imagine that there are some neural network architectures

with arcane topologies that implement that generative model.

And so as we build these rigorous bridges,

we see more and more with the same visual input.

And so that is really advancing the field.

And also, again, could not have been said a year ago.

Is it building a bridge to everywhere?

Or is it moving in that direction?

Let's go to the reflections.

Oh, okay.

If we built bridges to everywhere,

it's like paved paradise, put up a parking lot.

You know, we wanted people to be able to drive all over paradise

and park wherever they wanted.

So we paved it, we put up a parking lot.

I don't think that's what's happening.

Okay.

What is happening?

Well, that's, I mean, again,

I get the ambition of trying to take some statistical models

and see where they might be inserted

in things that are going out and doing the exploration.

See that.

And my, where am I,

because I,

my,

my continuing

attention to active inference

is because I believe it is a,
a way
when you're ready,
it is a way
of generating
way more insight,
way more aha moments
than
what I think is a
kind of unidirectional approach
or a hierarchical approach
that says,
mass,
all this information,
put it in,
synthesize it into your own,
synthesize it,
synthesize it
and feed it back to a,
a board or a review committee
or,
or something like that.

I understand
how that works
and why that
serves a certain
agenda
or,
or,
or intent.

I think what active inference does
is it opens
things up
so that
you do not have
a ladder to climb
or
at least they haven't invented

a ladder to everywhere.

I think that's

what it does

is it,

it creates options

and possibilities

and sees things

as

it could be this,

it could be that.

So from a humanist

standpoint,

um,

it just

makes available,

provides affordances.

That's

kind of

at the core

of the work

that's being done.

What,

it,

it,

it,

it enables

the use of things

like counterfactuals,

why that matters,

why

it's just as important

to ask what if

as it is to ask what is.

These are all things

that I think

don't even have to be stuck

in

narrative spaces.

I think it's,
it's
what you now
can choose
to try
and make some
sense out of.
And I think a lot
of times
in order for people
to become specialists
and experts
at something,
they have to really
focus their attention
on something.
but I think
what active inference
says is
the more times
you vary
your attention
and change
your attentional space,
i.e.
make yourself
available
to ideas
and phenomena
that you
otherwise would
ignore
because you
think
it's,
it doesn't really
matter in this
particular moment,

the more
ahas,
the more insights
you yourself
gain.
So,
I don't know
that we're paving
over anything.
I don't think
that the bridge
has to be something
positivist
and material.
In fact,
as we
talked about
at that very
second set
of live streams
this year,
is the thing,
is the hidden
state
hidden because
of a shroud,
something that's
material,
something that
blinds us
to the situation,
or is it
simply something
that we just
haven't
given our
attention to
over yet?

It's,
do I need,
do I need
to
build a bridge
across the
Grand Canyon
or is
seeing across
it to the
other side
enough?
That's what
active inference
asks us to
explore.
I think if
people want
to come
and find
an opportunity
to socialize
with other
people around
references,
there's work
to do with
that around
stuff that
Bajan has
written about
and stuff that
Carl has
written about,
and those are
fantastic exercises,
but I think
there's more

to it than
that,
and I think
the everywhere
part is
something that
again,
it gets
reinforced when
you're putting
on a live
stream that
you put in
hours and
hours and
hours of
preparation on
and you
don't know
whether or
not the
person who
actually was
the author
of the work
is going to
be able to
show up or
not because
these are
busy people
and then
they do.
That's an
active inference
moment.
That's a
kind of a

dedication to
a philosophy
and I think
that's where
Carl said
there's a
lot of
stuff that
still has
to be
worked out
by the
philosophers
while I
continue along
with Thomas
and Dalton
to try to
work out
what the
formalisms
and the
operations
are.
So I
think an
eye on
each is
kind of
where we
are today.
There's
probably less
room for
people to
say no
it's one
or the

other.

I think

there's a

lot more

room to

say well

if we put

this and

this and

this together

I wonder

what kind

of smoothie

that derives.

That's my

sense of

where we

are at

the start

of 2023.

Yes.

It was

like in

the beginning

of the

year if

only we

could just

pin it

down.

Is

active

inference

realist

or is

active

inference

this way?

If we
could just
get it
on the
table
on
Procrusty's
bed
then
I would
be able
to approach
it.
I'm too
busy right
now.
I'm too
overwhelmed
right now.
It's too
complex
right now.
It's too
ambiguous
right now.
It's changing
too fast
right now.
If you
could just
stabilize
it in
this one
dimension
then I'll
meet you
and I'll
be ready

to talk.
And out
of that
environment
the doorway
was
dialogue.
and getting
different
perspectives
together
beginner
intermediate
and beyond.
Today
it is
becoming
a growing
ecosystem
of
education
research
applications
and service
where
it's not
just that
there can
be the
coexistence
of different
perspectives
perspectives.
It's
that there
are,
there must
be,

and we
love that.
It seems
a bit
counterintuitive
but if there
is a nature
part to
this,
one of the
things we
might discover
at the end
of 2023
that there's
more pockets
of feral
and wild
behavior and
activity
as opposed
to less.
That the
more times
people come
and say,
is there a
packaged,
is there a
Disneyland
version of
this that I
can use as
my entry
point,
perhaps what
they'll discover
is there's

an opportunity.

when you

hit your

ball out

of bounds

in Costa

Rica,

there's

signs that

says,

do not

find your

ball.

There are

scorpions,

there are

snakes,

and there

are crocodiles.

Crocodiles

don't just

inhabit the

water areas.

They literally

hang out in

the forest.

So you

respect that.

Unless you're

going to

properly protect

your nine

iron ain't

going to

save you,

is my

point.

But there's

a piece of
this that's
highly groomed,
highly formalized,
and I think
there's also a
piece of
this that
from a
behavioral
active
standpoint,
we might
find a year
from now,
is a lot
more enabling,
is a lot
more feral,
is a lot
more who
wants to
go into
those marginal
areas and
really poke
around a
little bit.

And I
think,
again,
what people
might find
is that
active
inference
isn't a
shield,

but it
is an
anchor.
It provides
the
organizational,
structural,
and content-related
stabilities
that people can
anchor to,
so that they
can then go
out and play
with those
edges, or the
non-edges.
they can
function in
ways that
they probably
didn't imagine
if they
believed that
active inference
was a ladder
to something
specific.
So,
again,
there's a
creative side
of this
that I think
a year from
now,
again,
I know that

there's a bunch
of people in
business that
want to do
capture work
around this,
and there's a
bunch of
scientists that
want to get
confirmation,
and I get
that.

I get what
their goals
are, and I
respect them,
but I also
think there's a
creativity piece
to this that's
just percolating
below the
surface, and
that when it's
given a chance
to run its
course, it
could be
amazing in
terms of what
people generate.

It made me
think a
shield is not
a sarcophagus,
an anchor
is not a

pillar driven

into the

ocean floor.

Even

something that

is stabilizing

and defensive

must have

a loose

tethering

and an

optimal grasp,

and if

it's better

to be

stabilized,

why not

just put the

pillar down?

That is one

direction of

thought.

it may be

valid locally

or at a

mesoscale

situationally.

We're not

looking to

resolve these

questions in the

abstract.

There may be a

case where you

do want to get

inside the

iron sarcophagus

and hide,

but when
we are
out there
figuring
and balancing
that
with
figuring it
out,
we need to
also
see those
two in
conversation
or in
a dialectic
so it's not
like we can
just figure it
out such
that we're
going to
de-risk
and achieve
out there.
It's not
the work
itself.
It isn't
even,
interestingly
enough,
the exact
preparation for
the work
itself out
there.
But also,

if we're
only out
there,
so much
happens
that we
lose valuable
opportunities for
pre-reflexion and
reflection,
and we lose
those opportunities
to figure things
out while
the iron
is hot,
and so
we grow
and we act
and we serve
less than we
could,
which again
may not be a
bad thing,
and maybe
no lives are
lost for
it.

But
how to
read and
speak between
the lines
and make
sure that the
Active Inference
Institute

is accessible
to all
language and
mathematical
backgrounds
as our
usage of
the Active
Inference
Ontology
and the
formalisms
becomes
increasingly
sophisticated
as demands
from industry,
from
professionalized
research,
and all of
these other
sectors become
increasingly
intense as
well.

How will
we be able
to provide
services to
our community
and niche
that neither
exactly square
with what
their priors
are?

Because if it

did, they
could have
done it
themselves.

Nor do we
want to come
from a
different
wavelength or
planet where
the material
are failing
to make
impact because
they're not
accessible,
applicable,
and rigorous.

Here's an
interesting thing
to think
about as
one of the
people who
spent countless
hours growing
this idea.

I don't think
back in January
of this year
you could have
taken the
words from
Majid where
he was talking
about partial
encapsulation and
envisioned a

direct application
to the idea
that the
active inference
lab had to
be only
partially
encapsulated in
order for it
to morph into
the active
inference
institute.

These are the
things that we
upon reflection
can look back
and go,
well, we were
talking about it
as an idea,
but here's an
example of where
it actually
became true.

And we
couldn't have,
I didn't know
at the time
that the
active inference
lab was going
to turn into
the institute.

That's what
actively inferring
allows for.

That's what's

really fascinating.
And again,
I'll say this,
if you really
want to learn,
go through an
exercise like you
and I just went
through today.
Because it gives
you the opportunity
to say,
oh, by the way,
if you want some
real evidence
without pre-staging
the idea that I
want to prove
this,
I want to prove
that there's a
partial encapsulation,
just live it out.
Migrate through it.
Go through that
space-time,
come out the
other side,
look back on it
and go,
wow,
I guess what we
were talking
about 11
months and 26
days ago,
in fact,
could turn out

to be pretty
prescient.
Because there's
no way we could
become an
institute today
if we didn't
leave a gap
open for that
possibility.
So it's just
really interesting.
Again,
if you want
active inference
to be a
repeat of what
you have already
been habituated
to,
you can make
it that,
or you can
stick around
and kind of
go,
holy mackerel,
it wasn't just
in theory,
in fact,
it materialized
in the ways
that the theory
said it could.
not would,
and that's
the bridge
to everywhere.

Well,
we put up
our canonical
final slide.
Thank you,
Dean,
for this stream
and for all
of the others
and everyone
who came
on.
If you're
listening this
far,
thank you
for your
regime of
attention.
We are
looking to
make 2023
and years
beyond it
better.
And this
was a great
year,
so it's no
slight on
where we've
been.
it's just
a spiral
of development.
Any
dimension
or divergence

from your
highest
preference
is either
feedback you
can provide
the institute
or
engagement
that you
can yourself
make happen.

You want to
see a certain
kind of stream
happen?

You want to
see another
language other
than English?

You want to
see a different
kind of educational
material become
accessible?

either
provided us
feedback or
let's figure
out how to
make it
happen with
your
engagements.

We'll go
out there
anytime.

Wow.

Well,
no word
can close
the year,
so we'll
just leave
it at that.
Thank you,
Dean.
Thank you all.
All right.
Till next time,
my friend.
Till next time,
act and
first serve.
All right.
Ciao.
to

Thank you.